

Inorganic Chemistry Practical 2: Qualitative Analysis — Cation Separation & Identification

Systematic Diagnostic Precipitation, Amphoteric Differentiation, and Complex Ion Formation

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Inorganic Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Identification of Three Unknown Metal Cations: FA 1, FA 2, and FA 3**Cambridge Syllabus Reference: 9701/31/O/N/23/Q3 · [20 Marks]**

Qualitative Analysis Instructions:

You are provided with three aqueous solutions labeled FA 1, FA 2, and FA 3.

Each solution contains one of the following cations: Al^{3+} , Zn^{2+} , or Mg^{2+} .

Reagents Provided:

- Aqueous sodium hydroxide, NaOH(aq) , 2.0 mol dm^{-3}
- Aqueous ammonia, $\text{NH}_3(\text{aq})$, 2.0 mol dm^{-3}
- Dilute nitric acid, HNO_3 , 1.0 mol dm^{-3}
- Red and blue litmus papers.

Experimental Instructions:

Perform the specified diagnostic tests on 1 cm depth portions of each solution in clean test-tubes.

Record your observations in the table below. You should refer to the Qualitative Analysis Notes on pages 7–8.

(a) Carry out the following tests and record your observations in the table below. Include states, precipitate colors, and behavior in excess reagent. [7]

Test	Observations	Deductions
FA 1: (i) Add aqueous NaOH dropwise, then in excess. (ii) Add aqueous NH_3 dropwise, then in excess.		
FA 2: (i) Add aqueous NaOH dropwise, then in excess. (ii) Add aqueous NH_3 dropwise, then in excess.		
FA 3: (i) Add aqueous NaOH dropwise, then in excess. (ii) Add aqueous NH_3 dropwise, then in excess.		
Test for NH_4^+ : To 1 cm depth of FA 1, FA 2, and FA 3 in separate test-tubes, add 2 cm depth of NaOH(aq) and warm gently.		

(b) From your observations in (a), identify the cation present in each solution and justify your deduction. [6]

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(c) Write an ionic equation, with state symbols, for the precipitation reaction that occurs when dilute NaOH is added to FA 3. [2]

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(d) The precipitate formed initially with FA 1 dissolves in excess aqueous sodium hydroxide. Write a balanced chemical or ionic equation to account for this reaction and name the complex ion formed. [3]

(e) Explain why the precipitate formed by FA 2 dissolves in excess aqueous ammonia, whereas the precipitate formed by FA 1 does not. [2]

Question 2: Identification of Transition Metal Cations and Variable Oxidation States: FA 4, FA 5, and FA 6**Cambridge Syllabus Reference: 9701/35/M/J/22/Q3 · [20 Marks]****Qualitative Analysis Instructions:**

You are provided with three solutions: FA 4, FA 5, and FA 6.

Each solution contains one of the following transition metal cations: Cu^{2+} , Fe^{2+} , or Cr^{3+} .

Reagents Provided:

- Aqueous sodium hydroxide, $\text{NaOH}(\text{aq})$
- Aqueous ammonia, $\text{NH}_3(\text{aq})$
- Aqueous hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$
- Aqueous potassium iodide, $\text{KI}(\text{aq})$, and starch indicator solution.

Safety Advice:

Hydrogen peroxide is an oxidizing agent. In case of skin contact, rinse immediately with cold water.

(a) Carry out the specified tests on FA 4, FA 5, and FA 6 and complete the observation table below. [7]

Test	Observations	Deductions
FA 4: (i) Add $\text{NaOH}(\text{aq})$ dropwise, then in excess. (ii) Add $\text{NH}_3(\text{aq})$ dropwise, then in excess. (iii) Add aqueous KI .		
FA 5: (i) Add $\text{NaOH}(\text{aq})$ dropwise, then in excess. (ii) Leave to stand exposed to air for 5 minutes. (iii) To a fresh portion, add NaOH followed by $\text{H}_2\text{O}_2(\text{aq})$.		
FA 6: (i) Add $\text{NaOH}(\text{aq})$ dropwise, then in excess. (ii) Add $\text{NH}_3(\text{aq})$ dropwise, then in excess. (iii) To the solution from (i), add H_2O_2 and warm.		

(b) Identify the cations present in FA 4, FA 5, and FA 6. [3]

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(c) Write the formula of the copper species responsible for the deep blue solution formed when excess aqueous ammonia is added to FA 4. Identify the type of reaction and bonding involved. [3]

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(d) Explain the color change observed when the green precipitate formed by FA 5 is allowed to stand in air, and write an ionic equation for this aerial oxidation. [3]

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(e) In test (iii) for FA 6, explain the oxidation state change that occurs when the alkaline chromium solution is warmed with hydrogen peroxide. [4]

Qualitative Analysis Notes

Tables for the reactions of aqueous cations, anions, and tests for gases (Cambridge 9701 Syllabus Reference)

1. Reactions of Aqueous Cations

Cation	Reaction with NaOH(aq)	Reaction with NH ₃ (aq)
Aluminium, Al ³⁺	White ppt., soluble in excess giving a colorless solution	White ppt., insoluble in excess
Ammonium, NH ₄ ⁺	Ammonia gas produced on warming (turns damp red litmus blue)	No reaction
Barium, Ba ²⁺	Faint white ppt. or no ppt. (high [Ba ²⁺])	No ppt.
Calcium, Ca ²⁺	White ppt. with high [Ca ²⁺], insoluble in excess	No ppt. with dilute solutions
Chromium(III), Cr ³⁺	Grey-green ppt., soluble in excess giving dark green solution	Grey-green ppt., insoluble in excess
Copper(II), Cu ²⁺	Pale blue ppt., insoluble in excess	Pale blue ppt., soluble in excess giving intense deep blue solution
Iron(II), Fe ²⁺	Green ppt., insoluble in excess; turns brown on standing near surface	Green ppt., insoluble in excess; turns brown on standing
Iron(III), Fe ³⁺	Red-brown ppt., insoluble in excess	Red-brown ppt., insoluble in excess
Magnesium, Mg ²⁺	White ppt., insoluble in excess	White ppt., insoluble in excess
Manganese(II), Mn ²⁺	Off-white ppt., insoluble in excess; rapidly darkens to brown	Off-white ppt., insoluble in excess; darkens to brown
Zinc, Zn ²⁺	White ppt., soluble in excess giving a colorless solution	White ppt., soluble in excess giving a colorless solution

2. Reactions of Aqueous Anions

Ion	Test	Observation
Carbonate, CO ₃ ²⁻	Add dilute acid	Effervescence; gas turns limewater milky (CO ₂)
Chloride, Cl ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	White ppt., soluble in dilute aqueous ammonia
Bromide, Br ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	Cream ppt., insoluble in dilute NH ₃ , soluble in conc. NH ₃
Iodide, I ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	Yellow ppt., insoluble in dilute and conc. aqueous NH ₃
Nitrate, NO ₃ ⁻ (aq)	Warm with aqueous NaOH, then add Al foil	Ammonia produced; pungent gas turns damp red litmus blue
Sulfate, SO ₄ ²⁻ (aq)	Add dilute HNO ₃ , then aqueous Ba(NO ₃) ₂ / BaCl ₂	Dense white ppt. (insoluble in dilute strong acids)
Sulfite, SO ₃ ²⁻ (aq)	Add dilute acid, warm gently and test gas	Gas produced (SO ₂) turns acidified K ₂ Cr ₂ O ₇ from orange to green

3. Tests for Gases

Gas	Test and Test Result
Ammonia, NH ₃	Turns damp red litmus paper blue; choking pungent smell
Carbon dioxide, CO ₂	Gives a white precipitate with limewater (calcium hydroxide solution turns milky)
Chlorine, Cl ₂	Bleaches damp litmus paper white; pale green choking gas
Hydrogen, H ₂	'Pops' with a lighted wooden splint
Oxygen, O ₂	Relights a glowing wooden splint
Sulfur dioxide, SO ₂	Turns acidified potassium dichromate(VI) paper from orange to green

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Inorganic Chemistry Practical 2: Qualitative Analysis – Cation Separation & Identification

Question 1: Identification of Three Unknown Metal Cations: FA 1, FA 2, and FA 3 [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	FA 1 with NaOH: White precipitate [1], soluble in excess giving a colorless solution [1]. FA 1 with NH ₃ : White precipitate [1], insoluble in excess [1]. FA 2 with NaOH: White precipitate, soluble in excess giving colorless solution; with NH ₃ : White precipitate, soluble in excess giving colorless solution [1]. FA 3 with NaOH: White precipitate, insoluble in excess; with NH ₃ : White precipitate, insoluble in excess (or faint white ppt) [1]. Warming with NaOH: No ammonia gas produced / no pungent gas / damp red litmus stays red (rules out NH ₄ ⁺) [1].	[7]
(b)	FA 1 is Aluminium, Al ³⁺ [1]: forms white ppt soluble in excess NaOH (amphoteric) but insoluble in excess NH ₃ [1]. FA 2 is Zinc, Zn ²⁺ [1]: forms white ppt soluble in both excess NaOH (amphoteric) and excess NH ₃ (complex formation) [1]. FA 3 is Magnesium, Mg ²⁺ [1]: forms white ppt insoluble in excess of both NaOH and NH ₃ [1].	[6]
(c)	Mg ²⁺ (aq) + 2OH ⁻ (aq) → Mg(OH) ₂ (s) [2] (1 mark for correct formulae and balance, 1 mark for state symbols).	[2]
(d)	Al(OH) ₃ (s) + OH ⁻ (aq) → [Al(OH) ₄] ⁻ (aq) (or [Al(OH) ₄ (H ₂ O) ₂] ⁻) [2]. Name: Tetrahydroxoaluminate(III) ion [1].	[3]
(e)	Zn(OH) ₂ reacts with ammonia molecules which act as ligands to form the soluble tetraamminezinc(II) complex ion, [Zn(NH ₃) ₄] ²⁺ (aq) [1]; NH ₃ is a weak base providing insufficient [OH ⁻] to dissolve amphoteric Al(OH) ₃ as [Al(OH) ₄] ⁻ , and NH ₃ cannot act as a ligand to form stable ammine complexes with Al ³⁺ [1].	[2]

Question 2: Identification of Transition Metal Cations and Variable Oxidation States: FA 4, FA 5, and FA 6 [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	FA 4: Pale blue ppt with NaOH, insoluble in excess [1]; Pale blue ppt with NH ₃ , dissolving in excess to give deep blue solution [1]; With KI: White/off-white ppt of CuI in a dark brown solution of I ₂ (turns blue-black with starch) [1]. FA 5: Green ppt with NaOH, insoluble in excess [1]; turns brown / red-brown at surface on standing [1]; with H ₂ O ₂ : rapidly oxidizes to red-brown Fe(OH) ₃ precipitate with effervescence [1]. FA 6: Grey-green ppt with NaOH, dissolving in excess to dark green solution [1]; grey-green ppt with NH ₃ , insoluble in excess; with alkaline H ₂ O ₂ : green solution turns bright yellow (chromate, CrO ₄ ²⁻) [1].	[7]
(b)	FA 4 is Copper(II), Cu ²⁺ [1]; FA 5 is Iron(II), Fe ²⁺ [1]; FA 6 is Chromium(III), Cr ³⁺ [1].	[3]
(c)	Formula: [Cu(NH ₃) ₄ (H ₂ O) ₂] ²⁺ (or [Cu(NH ₃) ₄] ²⁺) [1]. Type of reaction: Ligand displacement / ligand exchange / complex formation [1]. Bonding: Dative covalent (coordinate) bonding from lone pair on N of NH ₃ to Cu ²⁺ [1].	[3]
(d)	Iron(II) hydroxide, Fe(OH) ₂ (green), is oxidized by atmospheric dissolved oxygen to iron(III) hydroxide, Fe(OH) ₃ (red-brown) [1]. Equation: 4Fe(OH) ₂ (s) + O ₂ (g) + 2H ₂ O(l) → 4Fe(OH) ₃ (s) [2] (or 4Fe ²⁺ + O ₂ + 4H ⁺ / ionic representation).	[3]
(e)	Chromium is oxidized from +3 in [Cr(OH) ₆] ³⁻ / [Cr(OH) ₄] ⁻ [1] to +6 in the yellow chromate(VI) ion, CrO ₄ ²⁻ [1]. Hydrogen peroxide is reduced from oxidation state -1 (in H ₂ O ₂) to -2 (in H ₂ O / OH ⁻) [1]. Equation: 2[Cr(OH) ₄] ⁻ + 3H ₂ O ₂ + 2OH ⁻ → 2CrO ₄ ²⁻ + 8H ₂ O [1].	[4]